

CSA1606-DWDM

KNIME Application Development Activities

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Activity 1: Sales Analysis Dashboard

Problem Statement

A retail company wants to analyze its monthly sales.

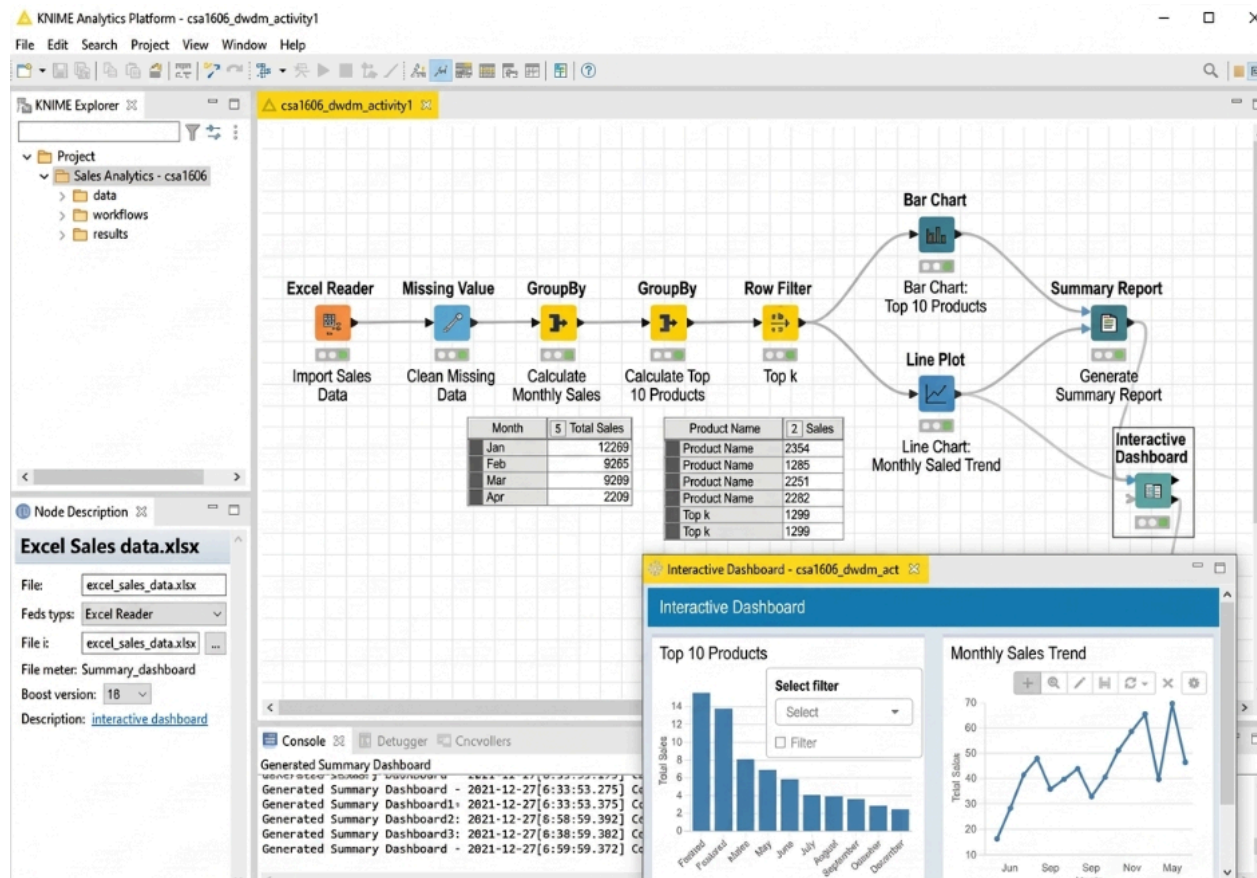
Task

Build a KNIME workflow that:

- Imports a sales dataset.
- Cleans missing values.
- Calculates monthly sales.
- Identifies the top 10 products.
- Creates bar and line charts.
- Generates a summary report.

Expected Outcome:

An interactive sales analysis workflow.



Activity 1: Sales Analysis Dashboard

Objective: Build an end-to-end KNIME pipeline that takes raw sales data and turns it into an interactive dashboard showing trends and top performers.

How it was built, node by node:

- **Excel Reader (Import Sales Data):** This is the entry point of the workflow. It reads the raw excel_sales_data.xlsx file into KNIME, pulling in columns like Month, Product Name, and Sales figures.
- **Missing Value (Clean Missing Data):** Real-world sales data almost always has gaps (empty cells due to unrecorded transactions). This node scans each column and fills or removes missing values – e.g., using mean/median substitution for numeric sales figures – so downstream calculations are not skewed.
- **GroupBy (Calculate Monthly Sales):** Groups all transaction rows by the Month column and aggregates the Sales column (using SUM), producing a clean monthly summary table (Jan, Feb, Mar... with Total Sales).
- **GroupBy (Calculate Top 10 Products):** A second, separate aggregation – this time grouping by Product Name and summing sales per product, so each product's total revenue contribution is calculated.
- **Row Filter (Top k):** Applied after sorting, this node trims the product-level table down to just the top 10 products by sales value, discarding the rest.
- **Bar Chart (Top 10 Products):** Takes the filtered top-10 table and renders it as a bar chart, visually comparing which products sold the most.
- **Line Plot (Monthly Sales Trend):** Takes the monthly aggregated table and plots it as a line graph, showing how sales rose or fell across months – useful for spotting seasonal trends.
- **Summary Report (Generate Summary Report):** Consolidates the key numbers (totals, top performers, trend direction) into a textual/tabular report node.
- **Interactive Dashboard:** The final composite view node that combines the bar chart and line chart into a single interactive dashboard with filters (e.g., a Select filter dropdown), letting the end user explore the data live rather than viewing static charts.

What this demonstrates: ETL (Extract–Transform–Load) fundamentals – import → clean → aggregate → filter → visualize – which is the standard shape of any BI/reporting pipeline.

Activity 2: Hospital Patient Risk Prediction

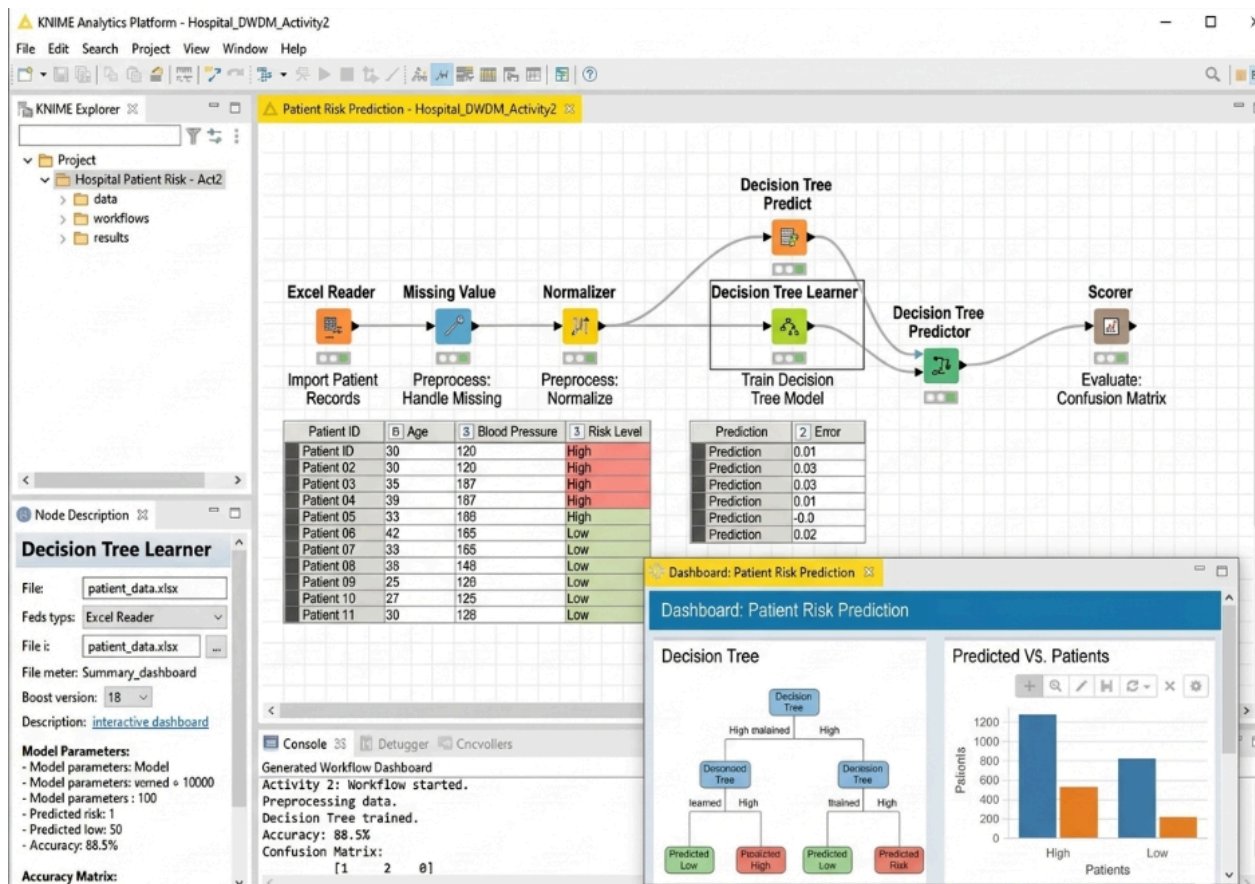
Problem Statement

A hospital wants to predict whether a patient is at high or low health risk.

Task

Develop a workflow to:

- Import patient records.
- Preprocess data.
- Train a Decision Tree model.
- Predict patient risk.
- Evaluate accuracy using a confusion matrix.



Activity 2: Hospital Patient Risk Prediction

Objective: Build a classification pipeline that predicts whether a patient is High or Low risk based on health attributes, and evaluate how good the prediction is.

How it was built, node by node:

- **Excel Reader (Import Patient Records):** Loads patient_data.xlsx containing columns like Patient ID, Age, Blood Pressure, and Risk Level (the target/label column).
- **Missing Value (Preprocess: Handle Missing):** Handles any blank cells in Age or Blood Pressure (e.g., a missing reading) so the model does not break or learn from bad data.
- **Normalizer (Preprocess: Normalize):** Scales numeric columns like Age and Blood Pressure to a common range (e.g., 0–1). This matters because Blood Pressure (~120–187) and Age (~25–42) are on very different scales, and normalization prevents one feature from dominating the model just because its numbers are bigger.
- **Decision Tree Learner (Train Decision Tree Model):** This is the core ML step. It takes the normalized, cleaned data and builds a decision tree that learns rules like "if Blood Pressure > X and Age > Y → High Risk." The node description panel shows parameters used (model count, boosting settings, etc.).
- **Decision Tree Predictor:** Applies the trained tree model to (typically held-out/test) patient records to generate a Prediction column – the model's guess of High or Low risk for each patient.
- **Scorer (Evaluate: Confusion Matrix):** Compares the model's predictions against the actual Risk Level labels and builds a confusion matrix, showing correct vs incorrect classifications (true positives, false positives, etc.) and computing overall accuracy (shown as 88.5% in the console log).
- **Dashboard (Decision Tree + Predicted vs Patients):** The final interactive view shows the trained decision tree structure visually (splitting logic) alongside a bar chart comparing predicted High vs Low counts against actual patient counts.

What this demonstrates: A supervised classification pipeline – import → preprocess/normalize → train →

predict → evaluate – which is the standard workflow for any ML classification task in KNIME.